

Physics 225, Spring 2026: Homework 7 - **Solutions**
Due 3/9/26 - Note **Monday is due day!**

All solutions must clearly show the steps or reasoning you used to arrive at your result. You will lose points for poorly written solutions or incorrect reasoning/explanations; answers given without explanation will not be graded.

Name:
Section:

1. **Invariant interval.** (60 points) Morin exercise 2.22 (p. 107). Please note that the speeds of B and C are both given with respect to the *ground*. You may use Lorentz transformations or the fundamental effects to find the coordinates, but you might find it useful practice to do it with Lorentz transformations.

- A 's frame: In the ground frame, the train has length $L/\gamma_{3/5} = 4L/5$. The relative speed between C and B in the ground frame is $4c/5 - 3c/5 = c/5$. The front of the train starts out $4L/5$ ahead of C , so the time it takes C to close this gap is $\Delta t_A = \frac{4L/5}{c/5} = 4L/c$. The distance between the events in the ground frame equals C 's speed times this time, which gives $\Delta x_A = (4c/5)(4L/c) = 16L/5$.
- B 's frame: The distance is $\Delta x_B = L$. From the velocity-addition formula, B sees C move with speed $\frac{4c/5 - 3c/5}{1 - (4/5)(3/5)} = 5c/13$. The time it takes C to travel the distance L is then $\Delta t_B = \frac{L}{5c/13} = \frac{13L}{5c}$.
- C 's frame: The distance is $\Delta x_C = 0$ because both events happen at the location of C . C sees the train move with speed $5c/13$, so it is length-contracted down to $L/\gamma_{5/13} = 12L/13$. The time it takes to pass C is therefore $\Delta t_C = \frac{12L/13}{5c/13} = \frac{12L}{5c}$.

The value of $c^2(\Delta t)^2 - (\Delta x)^2$ is the same in all three frames. Factoring out a common factor of L^2 , the arithmetic goes:

$$4^2 - (16/5)^2 = (13/5)^2 - 1^2 = (12/5)^2 - 0^2$$

2. **Minkowski diagrams.** (40 points) Morin exercise 2.25 (p. 108).

- (a) In the barn frame, the pole has length $L/\gamma_{3/5} = 4L/5$. During the process, the pole must travel a distance equal to the difference of the barn's and pole's lengths, which is $L - L/\gamma = L - 4L/5 = L/5$. It does this at speed $3c/5$, so the time elapsed in the barn frame is $\Delta t = \frac{L/5}{3c/5} = \frac{L}{3c}$. The distance between the events is simply $\Delta x = L$, because the events happen at the two ends of the barn. (The distance between events is *not* the $L/5$ distance the pole travels, since the latter is equivalent to the distance an individual end of the pole travels, while in this problem we care about events that involve the two different ends of the pole.)

- (b) In the pole frame, the order of events is indeed reversed from the order in the barn frame, because the length-contracted barn moves leftwards, meaning that the event involving the right end of the barn happens first. The two events are depicted in Fig. 1, with the left event chosen to be at the origin. From part (a), the ct coordinate of the right event is $c \times \frac{L}{3c} = L/3$. We want the events to be simultaneous in the S' frame, so we have chosen the x' axis such that both events lie on it (in other words, $t' = 0$ for the two events.) Since $\beta = \tan \theta$, and doing some trig on the diagram tells us that $\tan \theta = ct/x$, we have

$$\beta = \tan \theta = \frac{L/3}{L} = 1/3.$$

This tells us that the frame in which the events are simultaneous is moving at $v = c/3$ to the right with respect to the barn. A frame with speed less than $c/3$ will see the left event happen first, but a frame with speed $c/3 < v < c$ will see the right event happen first.

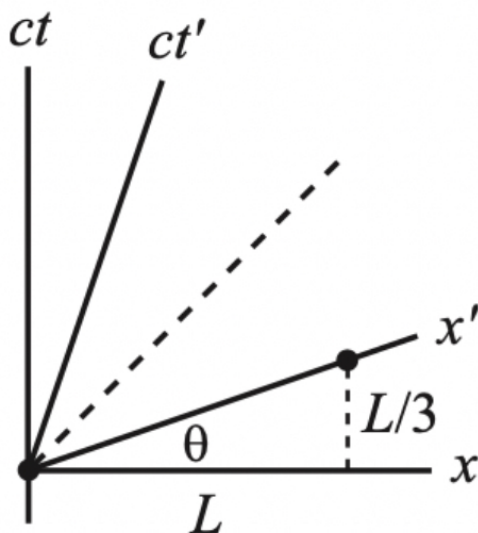


Figure 1: Minkowski diagram for 2.25 (b)